

DAA Practical 7

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Aim :- Implement Hamiltonian Cycle using Backtracking.

Problem Statement :-

The Smart City Transportation Department is designing a night-patrol route for security vehicles.

Each area of the city is represented as a vertex in a graph, and a road between two areas is represented as an edge.

The goal is to find a route that starts from the main headquarters (Area A), visits each area exactly once, and returns back to the headquarters — forming a Hamiltonian Cycle.

If such a route is not possible, display a suitable message.

1) Adjacency Matrix

```
A B C D E
A 0 1 1 0 1
B 1 0 1 1 0
C 1 1 0 1 0
D 0 1 1 0 1
E 1 0 0 1 0
```

2) Adjacency Matrix

```
T M S H C
T 0 1 1 0 1
M 1 0 1 1 0
S 1 1 0 1 1
H 0 1 1 0 1
C 1 0 1 1 0
```

Code :-

```
import sys

try:
    sys.stdout.reconfigure(encoding='utf-8')
    arrow = " → "
except:
    arrow = " -> "

def is_safe(v, graph, path, pos):
    if graph[path[pos - 1]][v] == 0:
        return False
    if v in path:
        return False
    return True

def hamiltonian_cycle_util(graph, path, pos):
    n = len(graph)
    if pos == n:
        return graph[path[pos - 1]][path[0]] == 1

    for v in range(1, n):
        if is_safe(v, graph, path, pos):
            path[pos] = v
            if hamiltonian_cycle_util(graph, path, pos + 1):
                return True
            path[pos] = -1
    return False

def hamiltonian_cycle(graph, vertices):
    n = len(graph)
    path = [-1] * n
    path[0] = 0

    if not hamiltonian_cycle_util(graph, path, 1):
```

```

        print("No Hamiltonian Cycle exists for the given city map.")
        return False

    print("Hamiltonian Cycle (Night Patrol Route):")
    for i in range(n):
        print(vertices[path[i]], end=arrow)
    print(vertices[path[0]])
    return True

print("=== Example 1: Areas A-E ===")
vertices1 = ['A', 'B', 'C', 'D', 'E']
graph1 = [
    [0, 1, 1, 0, 1],
    [1, 0, 1, 1, 0],
    [1, 1, 0, 1, 0],
    [0, 1, 1, 0, 1],
    [1, 0, 0, 1, 0]
]
hamiltonian_cycle(graph1, vertices1)

print("\n=== Example 2: Areas T-M-S-H-C ===")
vertices2 = ['T', 'M', 'S', 'H', 'C']
graph2 = [
    [0, 1, 1, 0, 1],
    [1, 0, 1, 1, 0],
    [1, 1, 0, 1, 1],
    [0, 1, 1, 0, 1],
    [1, 0, 1, 1, 0]
]
hamiltonian_cycle(graph2, vertices2)

```

Output :-

```
[Running] python -u "c:\Users\human\OneDrive\Desktop\Humanshu\pract7.py"
=== Example 1: Areas A-E ===
Hamiltonian Cycle (Night Patrol Route):
A → B → C → D → E → A

=== Example 2: Areas T-M-S-H-C ===
Hamiltonian Cycle (Night Patrol Route):
T → M → S → H → C → T

[Done] exited with code=0 in 0.131 seconds
```